

Question number	Answer	Additional guidance	Mark
*6(a)	Indicative content: • triplet codon system (D) • because (at least) 20 codes needed for the amino acids (E) • e.g. AAC is code for asparagine (x) • degenerate code (D) • therefore some amino acids have more than one code (E) • e.g. threonine can be coded for by ACA, ACC, ACG or ACT (x) • non-overlapping code (D) • so each base on DNA is used in only one triplet codon (E) • e.g. AAC AGA codes for two amino acids (x)	ACCEPT three bases code for one amino acid ACCEPT each amino acid has its own code ACCEPT there are more codes than necessary ACCEPT discrete	(6)
Level 1 : refers to triplet codon system, degenerative code or non-overlapping code but no examples or explanation given 1 mark = 1 out of 3 2 marks = 2 out of 3 or 1 out of 3 + a linked example or explanation Level 2 : refers to triplet codon system, degenerative code or non-overlapping code with either examples or explanation given 3 marks = at least 2 examples or 2 explanations or 1 of each 4 marks = at least 3 examples or 3 explanations or any combination of each Level 3 : refers to triplet codon system, degenerative code and non-overlapping code with examples and explanation given 5 marks = at least 4 examples or explanations or any combination of each 6 marks = at least 5 examples or explanations or any combination of each			

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6(b)(i)	methionine alanine cysteine proline isoleucine leucine	ACCEPT phonetic spelling / reasonable abbreviations / M A C P I L	(1)

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6(b)(ii)	An explanation that includes the following points: - it will have no effect (on the polypeptide) if the ninth base becomes a T as this still codes for {cysteine / same amino acid} (1) - will code for a stop codon if the ninth base becomes an A (1) - therefore the {protein / polypeptide} will be {shorter / not formed} (1) - will code for tryptophan if the ninth base becomes G (1) - this could change the bonding in the protein (1) - changing the {structure / activity} of the protein (1)	ACCEPT only two amino acids will join together ACCEPT even if tryptophan not given / given wrongly ACCEPT even if tryptophan not given / given wrongly	(5)